Cat Technologies

Cat Technologies combine the latest in construction technologies and services to help customers improve in four key areas: Equipment Management, Productivity, Safety, and Sustainability. Because every job site has its own unique challenges — whether running one machine, managing a fleet, or overseeing several job sites — Cat Technologies enable customers to combine technologies and services in ways that make the most sense for their business and realize significant gains in productivity, efficiency and profitability.

Cat Technologies offer the following technologies:

- Link
- Grade
- Compact
- Payload
- Detect

Cat Technologies offer the following services:

- Cat EMSolutions™
- Caterpillar Fleet Monitoring Center
- Productivity Services
- Job Site Solutions
- Safety Services

Cat Link Technology

Cat Link connects you to your assets (people or equipment) giving you access to essential information you need to know to run your business. Cat Link data can give you valuable insight into how your machine or fleet is performing so you can make timely, fact-based decisions that can boost job site efficiency and productivity.

Product Link

Product Link is the combination of GPS and telecommunications hardware used to communicate information about your assets so you can remotely monitor it. The GPS pinpoints the location while the integrated cellular or satellite radio enables data to be transmitted from many locations across the world. The data flows securely to the data servers at Caterpillar and then redistributed to you.

For Cat equipment, Product Link is designed as a factory standard or retrofit option for many models delivered to more than 50 countries. It is deeply integrated with engine, transmission and implement control systems to help you take the guesswork out of equipment management. Easy access to timely information can help you effectively manage your fleet and lower operating costs.

For other equipment brands, Product Link can communicate basic information — location, data derived from switches and switch inputs, and publicly available data on the other manufactures CAN bus.

The Product Link family provides you the ability to select the right fit options for your fleet operations.

VisionLink

VisionLink is a telematics software application that allows registered users to securely access information critical to their fleet and job site operations to:

- Manage assets using site boundaries, asset groups, shared asset views, etc.
- Manage scheduled maintenance, custom maintenance intervals, and major components.
- Monitor equipment health by monitoring fault codes, fluid analysis, tire monitoring and inspections.
- Maximize asset utilization and efficiency.
- Manage load counts.
- Create email or SMS alerts to remotely manage your fleet.
- Track payload data and fuel burn rates.
- Create reports
- Share data from or to other systems, i.e. ERP like SAP, accounting, or payroll.

Consult your dealer's Product Link specialist for more details or visit *cat.com/PL* or Link.

Basic Health or Advanced Health

Basic Health or Advanced Health is a fully integrated technology solution that uses sensors located throughout the machine to monitor crucial machine health, performance and productivity information. Basic Health or Advanced Health captures detailed, up-to-the minute data and displays it for the operator in real-time. The system alerts operators of abnormal machine conditions and provides instructions if action is needed. Data, such as historical trends, histograms, events and more can be transferred wirelessly or manually to the office for off-board analysis. Basic Health or Advanced Health software and the Health module within the Cat MineStar are predictive analysis tools used to analyze and interpret the data and provide reports that provide actionable information for informed decision making on machine performance, productivity issues, operations improvements, and fleet management. (Available data varies by machine model and type.)

Value of Basic Health or Advanced Health

- For the machine operator Basic Health or Advanced Health establishes two-way communication between the operator and the machine. Real-time machine information allows the operator to make informed decisions that directly affect safety, machine availability, and maximize productivity.
- For maintenance Basic Health or Advanced Health provides an in-depth view of operator and machine performance. This allows maintenance managers and technicians to maximize component life, reduce catastrophic failures, minimize unscheduled downtime and improve asset management.
- For production Basic Health or Advanced Health collects the information needed to monitor equipment usage, personnel performance, and productivity levels. Payload information can be used as an accounting tool, an indicator of cycle time efficiency and truck overloading or under loading.

Cat Grade Technology

Cat Grade combines digital design data, in-cab guidance, and automatic controls to enhance grading accuracy, reduce rework, and lower costs related to paving and earthmoving in rough, fine and finish grade applications.

Cat Grade increases increase productivity by moving material right the first time and achieving accurate grades without rework. These systems compute and track the location of the bucket or blade and compare this information with design parameters to guide to the operator to grade. In-cab displays with easy-to-use operator interface provide grade information in real time. Basic systems provide indicate-only information, while more enhanced systems can signal the machine hydraulics to move the blade to the desired design automatically. Cat Grade increases productivity, improves performance for less experienced operators, reduces operator fatigue, and enhances overall job site safety.

Cat Grade offers customers these advantages and cost-savings:

- Increase productivity
- Reduce fuel savings
- Reduce guesswork and costly rework
- Reduce survey costs
- Reduce staking, string lines and grade-checkers
- Increase material utilization
- Reduce owning and operating costs
- Reduce labor requirements and costs
- Finish jobs faster
- Work more confidently
- Extend the work day

Many aftermarket systems exist in the market today, but no system designed by someone other than the OEM can bring the advantages that integration makes possible.

Depending on the machine family and the specific application, many of the components necessary for automation are already present on a Cat machine. Cat Grade leverages these components and optimizes the design as only Caterpillar can. Whether it is a sensor, cylinder, display or software, Cat Grade integration is designed to reduce redundant components, protect sensors and allow productivity features to work together.

Cat Grade integrates traditional grade control with machine hardware and software in the factory to improve productivity, usability, reliability, job site safety and machine value. It is another example of Caterpillar innovation and technology leading the way for our customers to be more successful.

Cat Grade technology is comprised of the following:

- Cross Slope
- 3D
- Depth and Slope
- Slope Assist
- Grade and Slope

Cat Compact Technology

Cat Compact combines advanced compaction measurement, in-cab guidance and reporting capabilities to help you consistently meet compaction targets faster, more uniformly, and in fewer passes — reducing rework and material costs in both soil and asphalt applications.

Compaction is one of the most critical steps in the construction and road building process. Successful projects rely on achieving target compaction to stringent design specifications to ensure structural stability of the finished product.

Asphalt compaction technologies display mat temperature in the cab, indicating when and where operators need to work to quickly achieve consistent quality results. The system measures mat temperatures in real time, indicating the approach of tender zones and where the operator can work productively.

Soil compaction technologies give operators the information and instant feedback they need to achieve uniform results with maximum efficiency. Plus, it helps to identify hidden soil problems that can affect compaction quality.

Cat Compact offers customers these advantages and cost-savings:

- Increase productivity
- Reduce fuel savings
- Reduce guesswork and costly rework
- Increase material utilization
- Reduce owning and operating costs
- Reduce labor requirements and costs
- Provide Quality Control/Quality Assurance Documentation
- Finish jobs faster
- Work more confidently
- Extend the work day

Cat Compact provides information about the state of compaction to operators. The optional system comes with a choice of two different measurement technologies, Compaction Meter Value (CMV) and Machine Drive Power (MDP). CMV is an accelerometer-based system that displays a compaction value that indicates compaction quality. CMV is for smooth drum machines only and works well in granular applications and thick lifts. It is a Caterpillar exclusive technology that measures rolling resistance — with the vibratory system on or off — and correlates it with soil stiffness. MDP is for smooth and padfoot drums and works well in both cohesive or granular soils.

Optional 3D Mapping enables temperature or compaction measurements to be mapped to the precise location the operator is working, providing a real time view of progress. Mapping data is recorded to document compaction uniformity and job completion.

Cat Compact technology uses the following:

- Compaction Meter Value (CMV)
- Machine Drive Power (MDP)
- GNSS (SBAS)

Cat AccuGrade

Cat AccuGrade systems are dealer installed technologies that can be used to guide an operator to grade, manually using in-cab guidance, or automatically by controlling the blade movements to help operators get to grade faster, and more accurately. The systems use machine mounted sensors to calculate precise blade/tool location, slope and/or elevation information. The integrated electrohydraulic valve control module uses the information received from the sensors to control the machine's hydraulic system and automatically adjust the blade's cutting edge to maintain grade on select machines. Depending on the configuration, the operator can select which side of the blade/tool to control — right, left, or both sides. AccuGrade brings to the customer an increase in productivity by up to 40% while reducing their site costs dramatically.

Cat AccuGrade technologies include:

- Cross slope
- Sonic
- Site Reference
- Laser
- GNSS
- UTS

Cat Payload

Cat Payload enables accurate weight measurement of material being loaded or hauled. Payload information is shared with loader operators in real-time to improve productivity, reduce overloading and record both weights and number of loads per shift.

- Production Measurement
- Payload Estimator
- Truck Production Management System (TPMS)

Cat Production Measurement for Wheel Loaders, Excavators, and Articulated trucks is integrated from the factory and brings payload weighing to the cab to help operators work more productively and accurately; without rework. Operators can track load weights in real time on the in-cab monitor and know precisely how much material is in the bucket or truck. The system uses data from a series of onboard sensors that is processed by the onboard computer to calculate payload weight. Instant payload feedback through the display gives operators the confidence to work more efficiently and know when loads are filled to target capacity. Truck system also features external payload lights to indicate when the load is full.

Payload Estimator for Scrapers uses bowl lift cylinder pressure during the loaded haul segment to calculate payload. When working in manual operation, the system is optimized to work with Sequence Assist for increased productivity with less effort. Deep integration ensures reliable operation and accurate data (within ±5%).

Cat Payload maximizes the potential of the entire fleet:

- Enables operators to consistently fill to target payload
- Reduces under loading that reduces productivity, increases haul cycles, fuel usage, and operating costs
- Allows operators to track productivity such as weights and totals per shift
- Reduces overloading that can cause excessive wear on equipment and haul roads and lead to safety concerns

Truck Production Management System (TPMS) enhances truck and loader effectiveness to improve fleet productivity and reduce operating costs. TPMS offers external lights or an optional digital display to signal the loading tool operator when the proper load is reached. The on-board system provides a payload accuracy of ±5% and stores payload weight, cycle segment times, cycle segment distance and fuel, operator ID and actual clock time and date of each cycle.

Cat Detect Technology

Cat Detect combines safety and monitoring systems to enhance operator awareness. By expanding your view of the working environment around your equipment, you can improve the safety and productivity of your entire operation and keep your people and assets safe.

- Rear Vision cameras
- Work Area Vision System (WAVS)
- Tire Monitoring
- Machine Security System (MSS)

Rear vision cameras greatly enhance visibility behind the machine to help the operator work more productively. On many machines, the camera view is displayed through the standard display.

Work Area Vision System (WAVS) is used on larger equipment in applications where there are multiple machines, such as in a quarry. WAVS uses up to three cameras to provide full coverage in areas of limited visibility. Views can be set up by camera location (front, rear, side) or programmed to switch automatically based on direction of travel.

Tire Monitoring enables operators to monitor tire pressures and temperatures on the in-cab monitor. Alerts enable operators to take immediate action before a tire failure occurs and avoid unsafe operation.

Machine Security System (MSS) prevents unauthorized personnel from starting the machine by using a uniquely coded key that is programmed to a computer chip in the MSS module on-board the machine. The system can be set to prevent unauthorized use outside of normal working hours, and protect from vandalism or theft.

CAT MINESTAR

Cat MineStar is a comprehensive suite of technologies that allows you to see your entire operation at a glance, then drill down to the individual asset level as required. Its capability sets — Fleet, Terrain, Detect, Health and Command — enable you to define the size and scope of your management system based on the needs of your mining operation.

For more information on Mining Technology & Autonomy products, visit *cat.com/minestar.*

Fleet

With real-time machine tracking, assignment and productivity management, Fleet gives you a comprehensive overview of all your operations. Fleet enhances the management of all types of equipment operations and allows you to easily drill down for more detailed views and analysis. You can generate reporting on selectable groups of assets, equipment on a particular site or even individual machines.

Fleet works with data from all types of assets and equipment — including off-highway trucks, wheel loaders, motor graders, wheel dozers, shovels, light duty vehicles and equipment from other manufacturers — helping you reduce costs per ton, enhance productivity and boost overall site profitability.

Fleet is comprised of five capability packages, which can be purchased and configured based on a mine's particular needs.

- **Production** — Provides real-time visibility of production operations, delivers improved shovel loading performance and increases payload predictability.
- **Position & Material** — Monitors material movement and type, alerts operators and planners of misroutes to ensure material is moved to the proper location. Also monitors machine location for the entire fleet and incorporates playback function to analyze dump movement and haul road congestion.
- **Assignment & Optimization** — Schedules and assigns equipment, maximizes production and shovel utilization, minimizes truck wait time and manages shift changes and fueling.
- **Data Share** — Allows Fleet to share data made available via the licensed capability packages with other applications such as data reporting systems and position monitoring systems via an industry standard interface.

Fleet provides a proven solution suite based on a single trusted set of data for real-time KPI and for standard and ad-hoc reports. It provides the mine with information to:

- Identify and quantify performance improvement opportunities (within and post shift)
- Develop strategies to capture performance improvement initiatives
- Assign equipment and fleets for maximized production or achievement of material management goals. Capability scales from simple assignment to full truck assignment with linear programming to ensure maximal flexible loader, truck and material capacity utilization
- Blend materials in order to meet preparation plant quality, tonnage and timing requirements
- Track machines and materials to ensure correct delivery of materials from sources to planned sinks and to monitor equipment routing
- Manage operators (licensing, shift allocation and rostering)
- Manage equipment fluids and tires
- Track equipment productivity capability, consumption and variance
- Monitor equipment health including alarms and sensor channel monitoring, pre-start checklists
- Determine "what if" impacts of making specific changes to the product plan

Cost reduction of 10% and greater can be achieved and sustained using Fleet. Cost reductions are typically realized through reduced equipment, manning, lower fuel and service requirements, while achieving the same levels of productivity.

Terrain

Terrain enables high-precision management of drilling, grading and loading operations through the use of advanced guidance technology. It increases machine productivity and provides you real-time feedback for improved efficiency.

Along with providing detailed in-cab machine guidance and machine positional information for equipment operators, Terrain makes a wealth of data available to mine managers and site planners — including up-to-the-minute machine location and operational status, progress toward completion of work plans and more.

Terrain gives machine operators the real-time guidance they need to do their jobs more safely and efficiently, while providing mine site managers with timely information and advanced tools to help them increase mine productivity, output production and profitability.

Terrain capability packages include:

- **Productivity** — Productivity allows the mine to track and analyze machine utilization and productivity by machine type and operator. Reporting tools generate information on machine utilization, timelines, operator productivity and other parameters to help identify and correct operational inefficiencies. It also enables the assignment of job tasks to grading and loading tools. Operators can even request the creation and assignment of a task to another operator (such as clean-up a spill). The information about each task is tracked and stored for reporting purposes.
- **Position & Material** — The Position & Material capability package allows machines to share position and job status information both on-board and in the office. This knowledge helps reinforce safe operating practices when working in close proximity. The Position & Material capability package also enables machine-to-machine cut and fill status sharing within grading and loading applications in real-time including sharing cut/fill information from draglines to dozers.

- **Data Share** — Allows Terrain to share data made available via the licensed capability packages with other applications such as competitive fleet management systems, data reporting systems, and position monitoring systems via an industry standard interface.

Terrain for Drilling

Terrain for drilling is designed for installation on electric, hydraulic rotary blasthole, and articulated drills. It provides production and performance monitoring, strata recognition and GNSS guidance which replaces paint marks and paper patterns with centimeter accuracy and digital drill patterns. Provided as a field retrofit to machines already operating at mine sites, Terrain uses on-board computing integrated with sensors to monitor critical machine performance characteristics. System modules help the operator and site managers enhance drill performance and improve the drilling and blasting operation.

Production offers a graphical user interface to provide the operator with immediate feedback on drilling productivity and performance. The product minimizes operator input by an array of sensing hardware to detect:

- Hole depth
- Reaching target depth
- Drill pipe changes

Strata Recognition analyzes the monitored drilling variables in real-time, determining variability in the hole geology. The different strata horizons are presented on the display. The system provides useful and concise information from the start of drilling — not large amounts of raw data that typify traditional drill monitoring systems. A Blastability Index is determined by the Strata Module and approximates the hardness of the ground. The hole loading requirements and ore grindability predictions are then based on measured rock hardness enabling improved containment of explosive energy and more consistent fragmentation which impact downstream diggability, cycle times, fill factors and reduced costs for material movement. These enhance blending and optimized mill through put rates.

Combining Production with Strata Recognition logs:

- Bit rotary speed
- Penetration rate
- Depth
- Rotary torque or pressure
- Pull-down pressure
- Bailing air pressure

Drilling practice, efficiency and productivity can then be analyzed and assessed.

Guidance adds high precision GNSS to help precisely position a drill on a blast pattern without the need for surveying or staking. Guidance uses a moving map display that shows the 3D (Northing, Easting and Elevation) of the drill and drill bit relative to the designed position of the blastholes. Once the drill is positioned and leveled over a hole (hole parallelism), the system automatically determines collar elevation and then calculates the designed target depth. Guidance improves the drill's production and utilization, and the operator's ability to consistently drill to the plan. This leads to better rock fragmentation for easier loading. Since holes are drilled to the correct elevation leading to a flatter post-blast surface, the result is smoother pit floors. This helps eliminate rework, enhances the mobile equipment's performance and reduces its wear and tear.

Terrain for Grading & Loading

Terrain for grading & loading moves the material identification file and survey system into the machine, eliminating the need for survey stakes or pin flags. A touch screen monitor displays the location of pit boundaries, material type, bench height, and design grade, eliminating operator guesswork. With material types and locations displayed, ore identification and recovery are optimized.

The system is an ideal tool for mine planning, engineering, surveying, grade control and production monitoring applications. For example it can be used for:

- Haul road and bench construction and maintenance
- Production dozing
- Leach pad construction and maintenance
- Reclamation
- Task List Management
- Ore grade control and material identification
- Coal load out terminals

The system can be used on scrapers, loaders, dozers, shovels, motor graders, hydraulic excavators and track-type tractors.

Terrain features a mobile application for use in light vehicles. Mine supervisors can log in from a laptop or tablet to view progress updates from nearby machines, know the precise location of all Terrain-equipped machines, assign tasks and validate design plans without having to drive back to the mine office, improving the efficiency of your operations. Updates made on the mobile application are sent back to the office software and then communicated to the applicable machine onboard system in near real-time.

Detect

Operators often cannot see if another machine or vehicle is too close for safe operation. Detect helps alleviate this potential safety hazard and can also be configured to provide valuable information about site conditions and other assets working in the area.

At the most basic level, Detect enhances your operators' awareness of the immediate environment around their equipment. A simple touch screen display alerts operators when radars indicate that objects have entered critical areas near the machine. The system allows a quick visual check of these areas whenever the operator wants one.

Additional capability packages enable Detect to alert the operator to pre-programmed avoidance zones, known site hazards and speed limits. Positional information from Detect can also provide valuable feedback to central office systems and mine site managers.

Detect capability packages include:

- **Vision** — Vision offers multiple camera capabilities, allowing operators to select the view or views they need on the in-cab display to see what is happening close to their machines.
- **Object Detection** — Object Detection adds radars while reducing areas of limited visibility and increasing perimeter awareness. This robust system is scalable to site needs and machine types, providing optimal awareness around equipment.

Vision

Vision for surface equipment offers multiple camera capabilities, allowing operators to manually select the view or views they need on the in-cab display to see what is happening close to their machines. Certain Vision kits are configurable for manual or rotating camera views.

Object Detection

Object Detection is designed for machines ranging from large mining and quarry trucks to machines such as wheel loaders and motor graders. This robust system builds on Vision by adding radars to provide optimal awareness around the machine and notification when an object is detected. With both audible and visual indications, Object Detection helps prevent work area injuries caused by limited awareness. Using a combination of radars, cameras, and a high-resolution touch screen display, operators can view the areas immediately surrounding their machine, helping to prevent collisions and accidents.

Object Detection is highly integrated with the specific machine configuration to optimize radar and camera coverage. The system has been calibrated to provide appropriate fields of view and range. Unlike basic camera systems, Object Detection provides operators with audible and visual types of warnings that enable the operator to make informed decisions when moving or operating the machine. This system alerts the operator when an object is in close proximity so they can decide if action needs to be taken to avoid it.

Fatigue and Distraction Management

Fatigue, sleepiness and distraction impact each of us every day and losing focus on a mine site can have serious consequences. However, this risk can be managed and mitigated through a comprehensive Fatigue Risk Management System (FRMS) that incorporates all layers of protection against fatigue. These layers of protection include:

- Site fatigue assessments
- Training and education on managing fatigue for individuals, managers and supervisors
- Technology change management
- 24/7 monitoring
- Schedule/roster analysis and optimization

Caterpillar is the sole provider of a complete solution that brings visibility to risk factors never seen before and applies root cause data to a continuous improvement process that delivers sustainable culture change.

With specialized in-cab equipment that alerts an operator when a fatigue or distraction event is detected and data monitoring to identify risk trends, Caterpillar can help you prevent incidents in the moment and give line of sight to a host of factors that influence safety and operational performance. We also offer safety management consulting and training to help you build and sustain a culture that supports your fatigue management system.

Driver Safety System (DSS)

Fatigue and distraction are an inevitable force of nature. Their consequences can be costly, even fatal. Intervene before it's too late with a fatigue monitoring system. The Driver Safety System (DSS) is a non-intrusive, in-cab fatigue detection technology that instantly alerts operators the moment fatigue or distraction is identified.

Fatigue detection technology works by monitoring eye-closure duration and head pose. If the DSS detects a fatigue or distraction event the operator is immediately alerted through configurable in-vehicle seat vibration and/or audio alarm.

The fatigue or distraction event is sent to a 24-hour monitoring center to classify and confirm the event. Caterpillar experts will analyze the data and provide customized reporting with site-level recommendations. By cross-referencing fatigue and distraction events against available equipment data, Caterpillar can provide suggestions to improve operational efficiency.

Health

Cat® MineStar Health assists with improving the reliability of assets, reducing unplanned downtime, and preventing failures that can lead to lost productivity and costly machine repairs by providing a wide range of products and services that:

- Collect and transmit equipment data
- Monitor critical machine parameters
- Provide real-time alerts
- Identify operational trends and patterns
- Predict failures
- Provide repair recommendations

Health offers universal functionality that works with virtually any mine site asset including Surface and Underground, and Other Equipment Manufacturers (OEM) equipment.

Key Benefits of Health include:

- Lower maintenance costs
- Improved availability and reliability
- Saved component failures
- Extended component life

The MineStar Health portfolio set consists of the following:

- **Connectivity Enablers** – to collect, convert and transmit equipment data.
- **Applications** – to visualize, analyze and report health equipment data.

The Health portfolio also supports **Cat and Dealer provided condition monitoring services.**

Health connectivity products enable your equipment to collect and transmit machine electronic data into locally hosted or cloud-based applications.

Product Link Elite

Product Link Elite is the combination of GPS and telecommunications hardware used to securely collect and transmit machine electronic data into locally hosted or cloud-hosted applications for monitoring purposes.

Key Features:

- Dual data path – allowing data transfer to both the local server and the cloud simultaneously
- Cat or third-party radio compatibility – to suit site preferences
- Remote flash capability – for easy updating of firmware
- High-capacity memory storage – allowing for store-and-forward of data in the event of poor connectivity
- Ability to leverage Cat PL641 radio – for low precision GNSS needs

For Cat equipment, Product Link Elite is designed as a factory standard or as an aftermarket retrofit option for many models delivered to more than 50 countries.

For other equipment brands, Product Link Elite can communicate basic information — location, data derived from switches and switch inputs, and publicly available data on the other manufacturers' CAN bus.

Easy access to timely information can help you effectively manage your fleet and lower operating costs.

Product Link Elite Data Streaming

Product Link Elite Data Streaming is integrated with PLE and enables the transfer of over 150 machine health and operating data with up to 1 Hz sample rate.

The data transfer is completed via ethernet communication in an open and accessible format. Data is pushed out via an encrypted configuration into an off-board OSISoft® PI System™ to be consumed by third-party applications.

Data streaming has buffering capabilities to safeguard data in case of poor or no connectivity on site.

NOTE: A one-time Software Enabled Attachment (SEA) is required per machine to enable the Data Streaming functionality.

VIMS Forwarder

MineStar Health – VIMS Forwarder is a local site software application enabling data transfer from the local Dealer or Customer server to the Cat® Cloud at no cost.

It is designed to provide an easy and secure VIMS files processing and merging from Caterpillar and OEMs machines to enable MineStar Health Condition Monitoring applications and services, like Equipment Insights.

VIMS Forwarder is installed on a customer's computer (pc or laptop).

VIMS Converter

MineStar Health – VIMS Converter is a Windows Server based application allowing efficient conversion of VIMS data files from a proprietary format to a non-proprietary readable format (Comma Separated Value .csv) for ingestion into a system of choice.

VIMS Converter is compatible with all legacy and current VIMS files and systems:

- VIMS 68K with HIM or Product Link Elite
- VIMS ABL with HIM or Product Link Elite
- VIMS 3G
- Product Link Elite (ex-factory and aftermarket)

NOTE: Files are automatically sent to the Cat Cloud.

Health offers a variety of applications that visualize and report machine electronic data, such as VIMS and Product Link. These data visualization and reporting tools enable equipment reliability analysts, condition monitoring personnel, and/or maintenance managers to make better maintenance decisions.

VIMS Telemetry

The VIMS Telemetry port is a serial communication port available on many Cat machines at 1 Hz sample rate. VIMS Telemetry enables third-party access to Cat equipment health data by allowing a customer to receive about 92 machine operating parameters and events.

To activate VIMS Telemetry, Caterpillar requires customers and their third-party integrators to complete a Non-Disclosure Agreement (NDA) for each individual mine site before requesting:

- A Telemetry Port Events & Parameters document
- A VIMS Telemetry Port Description document

NOTE: A one-time Software Enabled Attachment (SEA) is required per machine to enable the VIMS Telemetry functionality.

VIMS Transmitter

MineStar Health – VIMS Transmitter enables the large scale and automated transmission of data from VIMS capable machines back to the Cat® Cloud. This process is followed by a data transfer into the Dealer back-office system via an Application Program Interface (API).

The API filters and transfers VIMS files only from machines with a VIMS Transmitter subscription activated in the Dealer Services Portal (DSP).

Health offers a variety of applications that visualize and report machine electronic data, such as VIMS and Product Link. These data visualization and reporting tools enable equipment reliability analysts, condition monitoring personnel, and/or maintenance managers to make better maintenance decisions.

Equipment Insights

MineStar Health – Equipment Insights is an off-board, cloud-based application that was developed to allow dealers and customers to view and analyze machine data from Cat and other brands of mining equipment.

Users have the option to visualize, analyze and report

- **Machine Electronic Data** (VIMS and Product Link) from Cat and select other OEMs
- **Fluids data** from Cat® S·O·S℠ or third-party fluid sampling providers
- **Inspection data** when Cat® Inspect is used

Users can customize or create their own dashboards by using more than 40 different widget options available. The system also provides options to generate automated e-mail reports.

Office

MineStar Health – Office is a server-based health application for customer sites and dealers.

It is a data visualization and reporting tool housed on the Dealer or Customer local server with various capabilities including but not limited to real-time event notification, real-time machine operating parameters visualization, remote log-in, as well as flexible and automated reporting system with more than 30 standard reports included.

Technician Toolbox

MineStar Health – Technician Toolbox is a PC software application that enables service technicians to perform machine configuration files to compatible payload and VIMS system parameters, and calibrate payload systems on the spot. Users can also trigger snapshots and activate VIMS dataloggers in order to gather critical machine health data.

In terms of machine troubleshooting, the VIMS Data can be downloaded and visualized in tables, graphs, and charts for one machine at a time through a customizable dashboard powered by 9 widgets.

PitLink

MineStar Health – PitLink is a client/server software application that forwards electronic data from a machine to the Caterpillar database (Cat Cloud) to enable a Cat application i.e. Equipment Insights, or a corporate MineStar Health – Office database.

PitLink also provides site personnel with the ability to perform live channel polling and visualize live machine events as they occur on the machine.

The MineStar Health product portfolio enables Cat Condition Monitoring Services for customers who desire a best-in-class predictive based condition monitoring program that delivers real value and cost reduction to an operation.

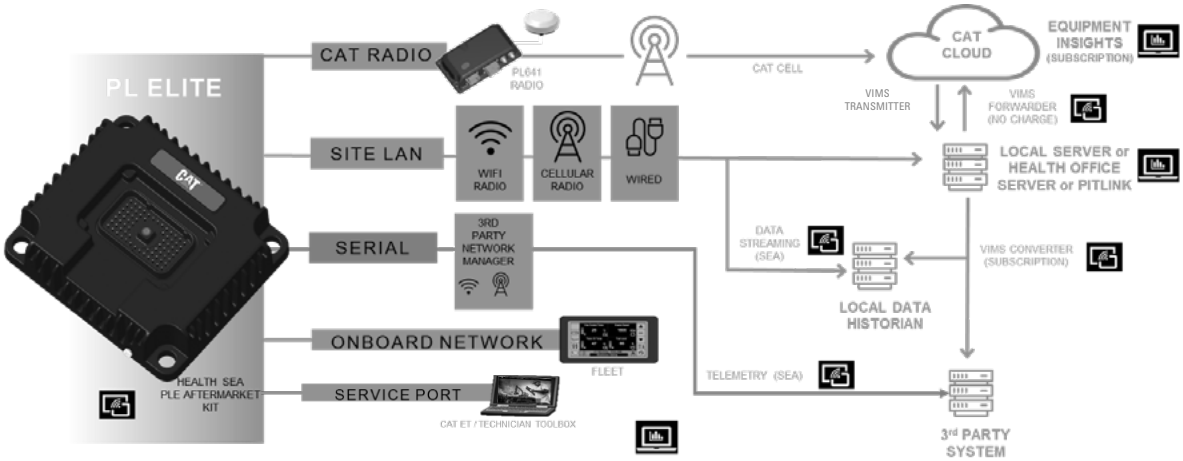
Condition Monitoring Service

Condition Monitoring Service is an offering that partners the maintenance and reliability organization of a mining operation with the strengths of the local Cat Dealer and MineStar Fleet Monitoring Center. The CM Service deploys a Predictive Maintenance Initiative through analytics with the goal of achieving the following objectives:

- Improve availability and reliability through predictive maintenance.
- Empower reliability engineers to improve maintenance planning.
- Extend machine life.
- Utilize machine condition monitoring data for analytics projects.

The service includes:

- Cat Condition Monitoring Advisor – Maintenance and health domain experts dedicated to the fleet and located in the MineStar Solutions' Fleet Monitoring Center to provide predictive maintenance recommendations.
- Predictive Analytical Software – Cloud hosted software designed specifically for the mining industry. Our solution uses advanced anomaly detection software resulting in highly confident correlation and causation determination. Our solution is supported by best in class analytical engines for condition monitoring data, including but not limited to electronic machine events, dataloggers, time series data, payloads, haul road conditions, work order histories, fluid sample results, and inspection results. Our Condition Monitoring Advisors work directly with a wide support group of Cat experts to include Data Scientists, Performance Engineers, and Product Group Engineers. These subject matter experts have knowledge of component design, component performance expectations, and have access to modeling and FMEA software for faster determination of root cause. Our software applications are maintained and supported within an agile development program, resulting in monthly quality improvements and feature introductions.
- Dynamic Health Data Visualization – Cloud hosted application providing real time visibility of machine, inspection, and fluids data at the mine site level, which is used by site reliability engineers and maintenance planners for PM planning, monitoring of critical machine parameters, and analysis of operational trends and patterns.
- Mentoring and Improvement Program – Resources deployed to site to support the local Maintenance Team make business process improvements to the maintenance programs.
- Data Connectivity – Hardware and software required to collect condition monitoring data from mining equipment at site and customer or dealer databases for transmission to the Fleet Monitoring Center.
- Application Program Interface – Real time data and information transfer between customer, dealer, and Caterpillar databases to affect the service.


PL ELITE
CAT RADIO
PL641 RADIO
CAT CELL
CAT CLOUD
EQUIPMENT INSIGHTS (SUBSCRIPTION)
VIMS TRANSMITTER
VIMS FORWARDER (NO CHARGE)
SITE LAN
WIFI RADIO
CELLULAR RADIO
WIRED
LOCAL SERVER or HEALTH OFFICE SERVER or PITLINK
SERIAL
3RD PARTY NETWORK MANAGER
DATA STREAMING (SEA)
VIMS CONVERTER (SUBSCRIPTION)
LOCAL DATA HISTORIAN
ONBOARD NETWORK
FLEET
TELEMETRY (SEA)
HEALTH SEA PLE AFTERMARKET KIT
SERVICE PORT
CAT ET / TECHNICIAN TOOLBOX
3rd PARTY SYSTEM

Command

Combining the Fleet, Terrain, Detect and Health capabilities of Cat MineStar, Command enables you to implement remote control, semi-autonomous or fully autonomous mining equipment systems for dozing and underground operations.

Taking advantage of proven technologies, as well as significant advances in remote sensing and guidance, Command helps you work more safely and productively in a wide range of harsh or challenging environments. And Command systems are proven to work safely and seamlessly with other mine site activities, equipment and personnel.

Command for Dozing

Command for dozing enables machine operation from a safe location with the operator removed from the cab. This system enhances operator safety by limiting exposure to full body vibration, dust and sound, and slips, trips and falls from machine ingress/egress. The machine can be operated by either an over-the-shoulder operator console or a seated operator station for longer term operations. The Console utilizes line of sight communications for near machine operation. With the addition of an on board vision system, the operator station can be located anywhere on-site or from an off-site command center. This system is integrated with engine, implement, and power train controls. Safety controls are built in which stop the machine in case of the loss of radio, transceiver, or ECM communications. The machine will also stop in the event the operator console is tipped. Additional protection features include the use of auto-brakes when in neutral and engine over-speed protection. Perhaps the most unique feature is the integration with the Terrain avoidance zone functionality, which prevents the machine from entering predefined avoidance zones.

Starting with a Remote Operator Station, a mine site can scale up to Semi-Autonomous dozing capability with the addition of our on-board planner system and MineStar Office. Command for dozing's semi-autonomous system is only available on the D11 (check your serial number for compatibility).

Semi-autonomous dozing allows an operator to sit comfortably and control multiple dozers from a Remote Operator Station while automating the dozer pivot push (late 2018) or push-to-edge processes for up to four machines simultaneously. This solution makes next-generation dozing a reality by allowing operation of dozers from a remote location either on the mine site or miles away, and reduces costs by tying multiple machines to a single operator and station.

Benefits of Command for dozing include:

- Enables control of machine from safe location when operating in hazardous conditions.
- Operator exposure to dust, noise and vibration can be minimized when operating remotely.
- Numerous emergency stop functions available in the following scenarios:
  - Operator Console is tipped.
  - Off-board transceiver loses power.
  - Radio communications are lost.
  - Communication to on-board transceiver or any electronic control module is lost.
- AutoBrake feature prevents machine from coasting while not being controlled.
- Avoidance Zone (optional) prevents a remotely controlled machine from entering predefined avoidance zones or 3-dimensional surfaces.
- AutoRetarder enables engine over-speed protection.
- Multiple emergency shutdown switches engaged in the cab, on the Operator Console, in the ground-level service center
- Meets ISO 15817 standards
- Meets AS/NZS 4240

Applications

- Machine recovery
- Unstable footings
- Aggressive ripping
- Misfires
- Stockpiles
- Bench slides
- Steep slopes

Command for Hauling

Command for hauling takes advantage of sophisticated technologies to enable Cat autonomous haul trucks to work safely and productively on busy mine sites. More than just an operator-free equipment system, Command for hauling is a complete, autonomous solution that delivers solid, bottom-line benefits for miners who need to work in challenging and remote locations. Highly advanced safety systems enable Cat autonomous haul trucks to operate reliably around other mining equipment, light vehicles and site employees.

Automation solves many problems faced by miners in today's competitive environment. With reserves being mined in more remote locations than ever, infrastructure requirements and labor shortages pose challenges that are difficult to overcome with manned mining operations. Automation resolves some of these challenges while making mines more safe, efficient and productive.

Benefits of Command for hauling include:

- Autonomous trucks work safely with other manned and light vehicles on the site.
- Multiple, redundant safety features ensure the system functions in a safe, predictable manner.
- High productivity through near continuous machine utilization.
- Reduction in process variability and better planning of maintenance and downtime improves operational efficiency.
- Supports your sustainability efforts by lessening infrastructure needs and operating the equipment as designed, reducing fuel burn, machine downtime and wear part replacement.

Command for Drilling

The Command for drilling automation solution offers a range of capabilities, so miners can configure and automate the drilling system to meet their budget and the needs of their site. Features and components serve as building blocks that allow customers to easily grow and add features and capabilities at their own pace. Command for drilling, a capability of the Cat MineStar technology suite, helps achieve excellent results through drill automation — which will ultimately reduce operating costs and improve productivity from predrill planning to blasting.

Semi-Autonomous Command for Drilling (S-ADS) automates the entire drilling cycle for one row, including autonomous tramming, without an operator in the cab. Flexible and simple, this system makes drilling automation easy to implement and more accessible for all kinds of operations. Autonomous drilling enables an operator to manage multiple drills across the mine from a Remote Operator Station (ROS) located on-site or off-site from a remote operating center.

Benefits of Command for drilling include:

- Improved accuracy
- Increased utilization
- Reduced operating costs
- Increased consistency
- Quality fragmentation
- Increased productivity
- Multiple redundant safety features ensure the system functions in a safe, predictable manner.
- Improved blasthole quality

Command for Underground

Developed out of the need to reduce human exposure to injury, the system removes the operator from dangerous situations and allows them to work in a more comfortable, ergonomic environment. The system uses technology to automate and enhance operations, by enabling full and semi-autonomous control of Cat LHD's. Command for underground will increase productivity and make a measurable impact on your mine's bottom line.

Command for underground consists of four major sub-systems that support the functionality of the system.

Remote Operator Station

The Operator Station allows machines to be operated from an ergonomically designed seat in a variety of locations. The operator can safely operate the machine from a mine control room or mobile office, either above or underground. This removes the operator from potentially dangerous environments in the mine drives underground. The Operator Station consists of three embedded PC displays, a Cat comfort seat and two joysticks, the other controlling the bucket.

Line of Sight Operator Console

With the line-of-sight Operator Console, operators have full control of basic machine functions and knowledge of critical machine diagnostics at the tip of their fingers — all at a safe distance from the machine. The ergonomic, durable, and modular over-the-shoulder harness allows for line-of-sight control of the LHD with communications on a 2.4 GHz frequency band. The remote control receiver is mounted on the machine and communicates with the operator console via the machine's electronic control module. The line of sight console can be used to control selected Load Haul Dumps equipped with a Cat line of sight receiver by switching the paired chip in the transmitter.

Machine Automation System

The Machine Automation System consists of the onboard hardware components that make the Command for underground system function. LADAR, cameras, lights, sensors, antennas, and control modules combine to create a system that provides safety and productivity for your underground mining operation.

Area Isolation System

Ensures that personnel cannot enter or equipment cannot leave the operations area while the machine is in autonomous mode. The operator has the ability to arm and disarm the system to compensate for changing business needs. A barrier control panel is located at each entry to the operations area. These are connected to barriers to ensure the area is secure. The status of each barrier control panel is reported to the programmable logic controller via the Local Area Radio Network (LARN). The programmable logic controller then determines whether the Operations Area should be armed.

Local Area Radio Network (LARN)

The LARN is a wireless Ethernet data network that enables communication between the machine and the Operator Station. The network uses the 802.11 b/g protocols and requires exclusive use of the 2.4 GHz RF spectrum. Signals from the Machine Automation System roam between LARN antennas as the machine moves within the operations area. The signals work primarily over line-of-sight, but can reach a short distance around corners. Video images and data are sent via the LARN.

Edge

MineStar Edge is a new technology platform that delivers more connected, integrated, scalable and intuitive products that extend further up and down the value chain. It creates an operational ecosystem that gives visibility to every aspect of an operation, allowing efficient execution of the mine plan and delivering optimal quality to the plant and to the surface. MineStar Edge makes it possible to measure, manage, analyze, and optimize the entire operation. And because it's delivered as a cloud-based, subscription managed application, it lowers costs and delivers a better user experience.

The best technology systems available today—when used to their full potential by personnel—deliver information that is about 80 to 90% accurate. MineStar Edge provides data that is 98% accurate. The platform makes use of Artificial Intelligence—data fusion and machine learning—to determine and communicate the accuracy of the information it is providing. Most data is automatically provided by the machines themselves, which eliminates the need for operator input and the risk of incorrect data entry. Near-real-time reporting is one of the most powerful features of MineStar Edge. Having access to the most up-to-date information allows mines to make quick adjustments to meet production goals rather than waiting until the end of the shift to determine if goals were realized. These decisions have an immediate impact on productivity.

MineStar Edge allows sites to start small and scale up to a full operational ecosystem. Users select an offering by role, function, or task—paying only for those functionalities they need. This model allows operations of all sizes to scale the technology in a cost effective and efficient manner. Because it is delivered as a cloud-based, subscription-managed application, MineStar Edge lowers network requirements as well as associated costs of deployment, service, and training. Mines are not required to invest capital in servers and database solutions that are sometimes required with mining technology solutions. The cloud-based platform also speeds the time from order to use, and allows automatic upgrades, updates, and fixes. In addition, sites have the option to have some applications hosted locally to ensure higher connectivity and less latency.

Equipment Tracking

Traditionally, sites have used pencil and paper to keep track of information on the majority of their equipment assets. Even if they leverage Production Recording technology to get data on the load/haul cycle, they have lacked access to the same information for non-production machines like water trucks, dozers, graders, drills, light vehicles, etc. MineStar Edge Equipment Tracking provides information on these mobile assets, no matter the manufacturer. It tells supervisors what the fleet is doing, who is operating which machine, and how they are spending their time. By replacing paper with accurate recording, supervisors can better understand utilization and make decisions about equipment investments and effectivity. Equipment Tracking provides insights that help mines better manage their assets, and as a result they deliver more value and mines enjoy a lower overall cost of ownership.

Equipment Tracking consistently provides a basic level of information on all assets, including data on locations and movements, velocity, payload, cycle times and fuel level, as well as Service Meter Readings and time utilization. Supervisors can view scheduled and unscheduled downtime along with scheduled and unscheduled operational stoppages. With fuel being one of a site's largest expenses, the ability to visualize and export fueling records is a key benefit of Equipment Tracking. This feature automatically measures fuel events—from when the machine was fueled to what was put in the tank and how long fueling took—allowing operations to manage the efficiencies of their fueling processes and fuel consumptions. It reduces or eliminates errors because data is collected automatically, and also lowers the cost of fuel.

Features and Benefits

- Gathers SMU data continuously from all assets. Data is no longer gathered manually through operators recording on paper or calling over the radio—eliminating incorrect entries.
- Delivers high-accuracy data recording, reducing downstream errors when the position is referenced for automatic collection of information.
- Provides the ability to replay shifts in great detail, helping sites answer Production Recording questions and providing information for incident investigation and material tracking investigation.
- Records the time that equipment is available, down, operating or stopped due to a nonoperating activity, providing accurate KPIs for use in time utilization models and leading to increased productivity and production.
- Automatically measures time when the equipment is not being productive and will attempt to classify with a machine-learning approach, reducing or eliminating missed or incorrect time measuring and giving operations information they can use to measure and manage their processes—specifically those that have production loss impacts.
- Helps sites measure and manage equipment downtime and classify it as scheduled or unscheduled, providing a key indication of the effectiveness of the site's maintenance and reliability programs, which can reduce downtime and lead to improved productivity.
- Allows operations to better manage and measure equipment operators, including allocating them to machines and tracking performance, and provides operators with KPIs about their personal performance—helping them improve and leading to a more consistent operation and reduced cost per ton.
- Helps operations improve equipment availability, reliability, and production by measuring, managing and ultimately reducing machine health events and making data easily accessible.

Production Recording

Every mining operation has opportunities hidden within every shift. But how do they find those opportunities if they're not accurately measuring what's happening during that shift? From payload to dig rates to operator breaks—every activity has an impact on productivity and an opportunity to be improved. MineStar Edge Production Recording helps sites find those opportunities by giving them visibility to the entire mining operation. When paired with Equipment Tracking, it delivers an accurate and automated near-real-time solution that measures and reports on every aspect of the load-haul-dump cycle without requiring any operator input. The result is a boost in productivity and a reduction in the overall operating costs of managed assets: mine, material, and machines. For many sites, traditional comprehensive fleet management systems can be too complex and cost-prohibitive, requiring significant investments in time and money to set up and maintain. Production Recording, however, is an easy-to-use subscription-based solution that delivers the key functionality all sites require: accurate, real-time production data.

Production Recording helps mines of all types and sizes improve the efficiency of their operations and increase their overall tons produced. The data it provides identifies opportunities, allows sites to make changes within the shift, and delivers insights on how those changes will impact production if implemented. The accuracy of the data gives personnel such as Pit Supervisors, Mine Managers, and Install Technicians the confidence they need to make quick, real-time decisions related to operational execution. Data can be accessed on a mobile tablet device as well as through a web-based application. A production dashboard provides information on the operation, including defined materials, active load and dump areas, active load and haul equipment, and active crushers. The dashboard enables viewing of hourly production metrics as well as cumulative shift-to-date production metrics for the site, material, load and dump areas, equipment, and the crusher. Beyond the shift, Production Recording allows sites to continually monitor and make operational decisions and implement training that will allow them take advantage of the opportunities they uncover.

Opportunities

- Missed production targets. Every mine has production goals such as total tons moved, dig rate, rate of material put through crusher, etc. All impact management's ability to measure progress toward production. Accurate information allows them to course-correct—in real time—to ensure the operation is on target.
- Payload compliance. Under-loading trucks means not getting maximum value from the truck or the cycle and impacts cost per ton. Over-loading can impact component life, boost maintenance costs, and increase unplanned downtime. Finding a balance equates to more effective machine utilization over the course of a shift.
- Shift changes/long operator breaks. The goal of an efficient mining operation is to effectively use all resources. Shift change and operator breaks, while necessary, must be managed to reduce the impact they have—keeping operators productive and reducing machine idle time.
- Misplaced loads. When material is dumped in the wrong location, mines take an immediate hit to their profitability. For example, if waste is dumped into the crusher or ore is dumped into the waste pile, the result is lost revenue. And if the blend is inaccurate, the finished product will result in less profit.

Features and Benefits:

- Monitors and provides highly accurate data on load and dump counts, tons moved and Bank Cost Per Meter/Bank Cost Per Yard.
- Provides the ability to track progress against plan in real time throughout the shift.
- Allows users to drill down to see the performance of entities such as individual machines, areas, routes, materials, and operators.
- Calculates and visualizes the expected production at the end of the shift for the site and individual entities including site, load and dump areas, routes, loading tools and trucks, and material.
- Enables sites to review production records from the point of view of the material transaction and movement from the load face to the dump, including the times, machines and operators involved.
- Delivers accurate transactional records that can be used as a source of truth for material movements.

TIRES

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SELECTION, APPLICATION, MAINTENANCE

Proper tire selection, application and maintenance continue to be the most important factors in earthmoving economics. Wheel tractors, loaders, scrapers, trucks, motor graders, etc. are earthmoving equipment whose productivity and payload unit cost may depend more on tire performance than any other factor.

Off-the-road tires must operate under a wide variety of conditions ranging from dry "potato dirt" through wet severe shot rock. Speed conditions vary from less than 1 mph average to 72 km/h (45 mph). Gradients may vary from 75% favorable to 30% adverse. Climatic conditions, operator skills, maintenance practices, etc. all may have a profound effect on tire life and unit costs.

Although one specific tire construction may be acceptable in a variety of applications, no one tire can meet all requirements on any one machine and perhaps not even one job. The many differences in tire requirements on earthmoving machines have resulted in a wide variety of tread and casing designs being made available. The optimum tire selection for a specific machine on a given job should be a joint decision between the user and tire supplier. Several tire manufacturers have technical and application representatives in the field for proper guidance in tire selection.

When job conditions change, it may be desirable to select a different tire configuration to meet the new requirements.